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GOALS

- * Efficient CPU Utilization
- * Optimal Memory Usage
- * High Throughput
- * Fast response time
- * Fairness and security

Memory Management

(for Process)

- * Allocate and Deallocate memory
- * Keep track of memory usage
- * Support Virtual memory
- * Ensure protection and isolation

Techniques

- * Paging
- * Segmentation
- * Swapping
- * Virtual memory

Interact with Each Component

- * processes request memory and I/O services
- * CPU schedules processes and handles interrupts
- * memory stores process data and I/O buffers
- * I/O system transfers data b/w devices and memory
- * All components communicate via system calls and interrupts

PROCESS

CONCEPT

Map

Process Management

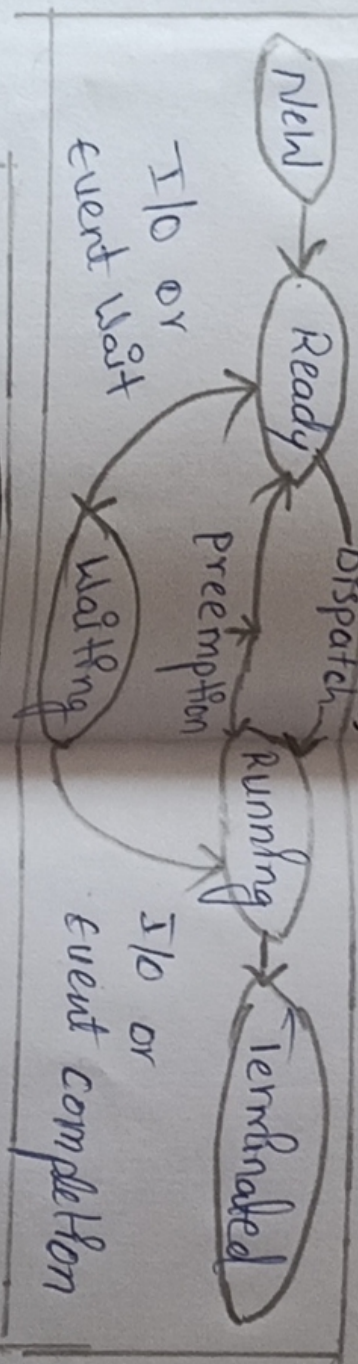
The process of overseeing and controlling the life cycle of processes in an operating system to ensure efficient execution of tasks.

Process Management

(for System)

- * Create and terminate processes
- * Suspend and resume processes
- * Maintain process states
- * provide process control blocks (PCB)

Process States



Interdependencies

Process management, memory management, and I/O systems are closely interrelated. They work together to ensure efficient system performance.

Process Management

Needs more memory to create and run processes

Memory Management

Allocate memory for processes and I/O operations

I/O Systems

provide data to/from processes via memory buffers

Legend

- * Memory Related
- * Process Related
- * I/O Related
- * Interrelation

85%
100%
60%
72%

Key Activities

- * process creation & termination
- * CPU scheduling
- * Synchronization
- * communication (IPC)
- * Deadlock handling

I/O System

(for processes and I/O)

- * Manage I/O devices
- * Provide buffer management
- * Handle device allocation
- * Ensure efficient I/O operation

Components

- * Device Drivers
- * Interrupt Handlers
- * I/O buffers
- * Spooling

Who Share System

- * Multiple processes share CPU, memory, and I/O devices
- * Sharing is managed by the OS to ensure fairness, protection and efficiency